

Mall Customer Analysis Report

Summary, graph, written analysis, and customer-level data

Customers

200

Average age

38.85

Average salary

60.56 k\$

Average spending

50.20

What the data shows

The dataset contains 200 customers with age, gender, annual income, and spending score fields.

Gender split: Female 112, Male 88.

Average salary is 60.56k, with values from 15.00k to 137.00k.

The largest rule-based segment is Balanced shoppers with 76 customers.

The next page visualizes each customer by age and salary, using color and shape to differentiate gender.

Files created by this program

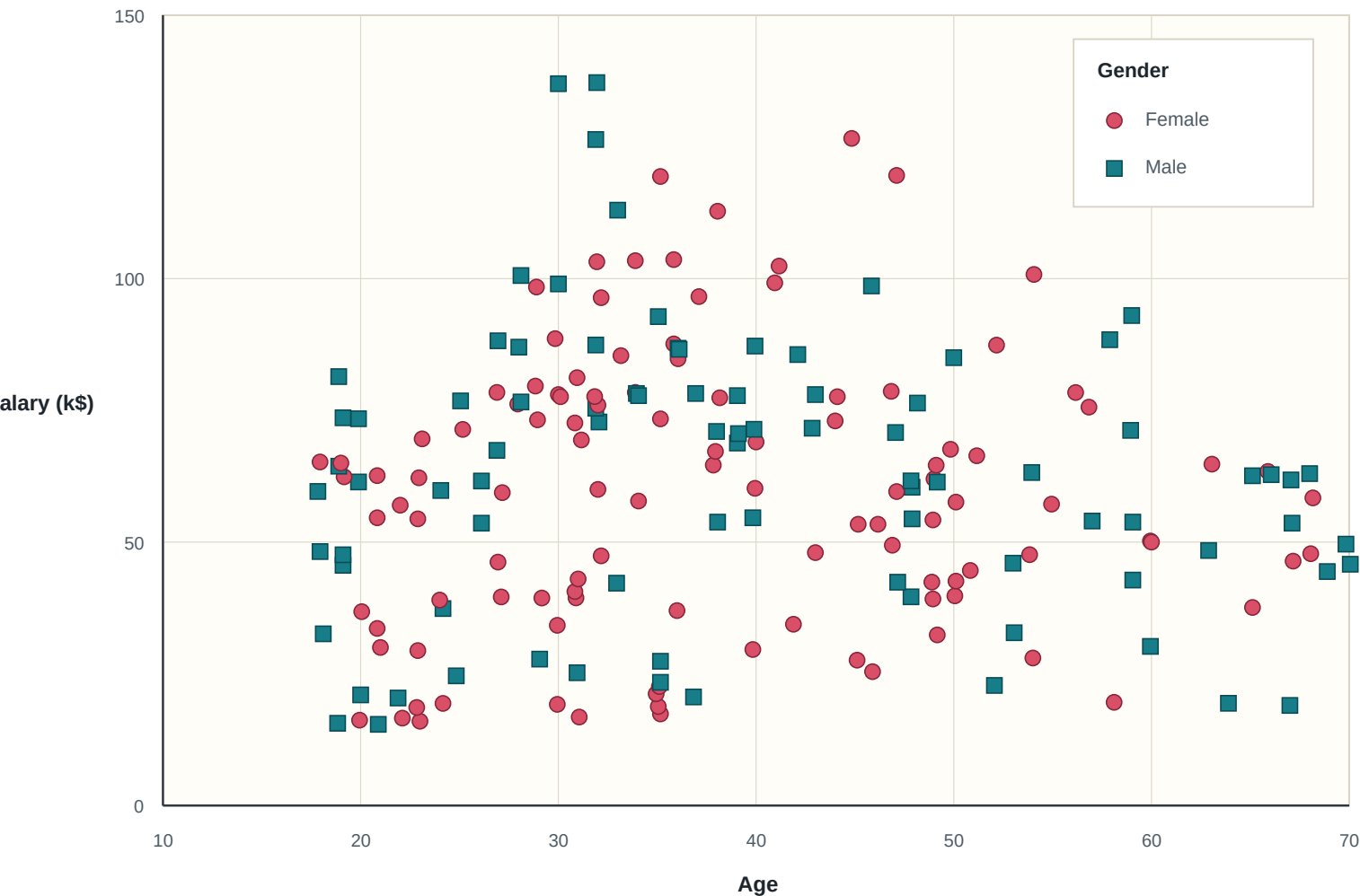
```
mall_customer_analysis_report.txt  
mall_customer_segments.csv  
mall_customer_age_gender_salary.svg  
mall_customer_analysis_report.pdf
```

Graph: Age, Gender, and Salary

Scatter plot plus average salary by age band

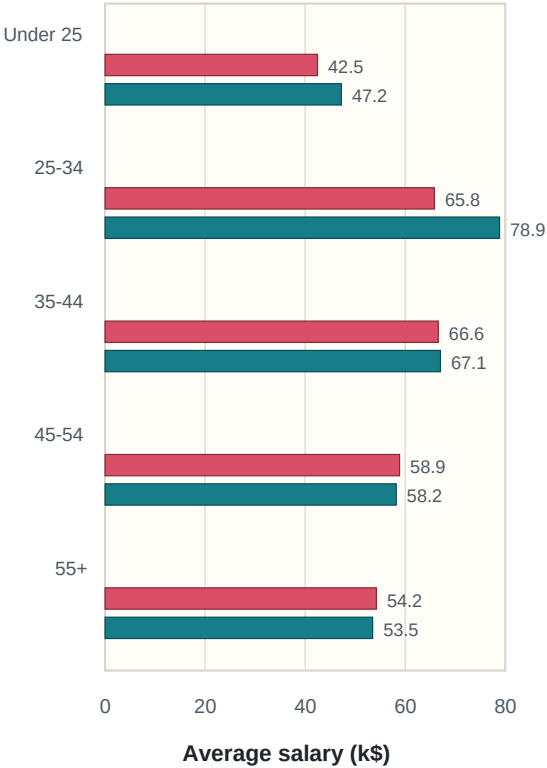
Each Customer: Age vs Salary

Annual income is measured in thousands of dollars.



Average Salary

By age band and gender



Detailed Analysis

Text version of the generated report

Mall Customer Data Analysis

=====

Total customers: 200

Column Summary

Age:

Count: 200

Missing: 0

Min/Q1/Median/Mean/Q3/Max: 18.00 / 28.75 / 36.00 / 38.85 / 49.00 / 70.00

Std dev: 13.93

Annual Income (k\$):

Count: 200

Missing: 0

Min/Q1/Median/Mean/Q3/Max: 15.00 / 41.50 / 61.50 / 60.56 / 78.00 / 137.00

Std dev: 26.20

Spending Score (1-100):

Count: 200

Missing: 0

Min/Q1/Median/Mean/Q3/Max: 1.00 / 34.75 / 50.00 / 50.20 / 73.00 / 99.00

Std dev: 25.76

Gender Distribution

Female 112 (56.0%) |#####.....|

Male 88 (44.0%) |#####.....|

Age Groups

Under 25 35 (17.5%) |#####.....|

25-34 54 (27.0%) |#####.....|

35-44 42 (21.0%) |#####.....|

45-54 39 (19.5%) |#####.....|

55+ 30 (15.0%) |#####.....|

Rule-Based Customer Segments

Balanced shoppers 76 (38.0%) |#####.....|

Premium targets 38 (19.0%) |#####.....|

High income, low spend 37 (18.5%) |#####.....|

Budget enthusiasts 23 (11.5%) |#####.....|

Low activity 22 (11.0%) |#####.....|

Active shoppers 4 (2.0%) |#.....|

Correlations

Age vs Annual Income (k\$): -0.012

Age vs Spending Score (1-100): -0.327

Annual Income (k\$) vs Spending Score (1-100): 0.010

Top 10 Customers By Spending Score

Customer 0012: Female, age 35, income 19k, spending 99

Customer 0020: Female, age 35, income 23k, spending 98

Customer 0146: Male, age 28, income 77k, spending 97

Customer 0186: Male, age 30, income 99k, spending 97

Customer 0128: Male, age 40, income 71k, spending 95

Customer 0168: Female, age 33, income 86k, spending 95

Customer 0008: Female, age 23, income 18k, spending 94

Customer 0142: Male, age 32, income 75k, spending 93

Customer 0164: Female, age 31, income 81k, spending 93

Customer 0034: Male, age 18, income 33k, spending 92

Customer Data

Rows 1-39 of 200

ID	Gender	Age	Salary	Spend	Age Group	Segment	Cluster
0001	Male	19	15	39	Under 25	Low activity	3
0002	Male	21	15	81	Under 25	Budget enthusiasts	5
0003	Female	20	16	6	Under 25	Low activity	3
0004	Female	23	16	77	Under 25	Budget enthusiasts	5
0005	Female	31	17	40	25-34	Balanced shoppers	3
0006	Female	22	17	76	Under 25	Budget enthusiasts	5
0007	Female	35	18	6	35-44	Low activity	3
0008	Female	23	18	94	Under 25	Budget enthusiasts	5
0009	Male	64	19	3	55+	Low activity	3
0010	Female	30	19	72	25-34	Budget enthusiasts	5
0011	Male	67	19	14	55+	Low activity	3
0012	Female	35	19	99	35-44	Budget enthusiasts	5
0013	Female	58	20	15	55+	Low activity	3
0014	Female	24	20	77	Under 25	Budget enthusiasts	5
0015	Male	37	20	13	35-44	Low activity	3
0016	Male	22	20	79	Under 25	Budget enthusiasts	5
0017	Female	35	21	35	35-44	Low activity	3
0018	Male	20	21	66	Under 25	Budget enthusiasts	5
0019	Male	52	23	29	45-54	Low activity	3
0020	Female	35	23	98	35-44	Budget enthusiasts	5
0021	Male	35	24	35	35-44	Low activity	3
0022	Male	25	24	73	25-34	Budget enthusiasts	5
0023	Female	46	25	5	45-54	Low activity	3
0024	Male	31	25	73	25-34	Budget enthusiasts	5
0025	Female	54	28	14	45-54	Low activity	3
0026	Male	29	28	82	25-34	Budget enthusiasts	5
0027	Female	45	28	32	45-54	Low activity	3
0028	Male	35	28	61	35-44	Budget enthusiasts	5
0029	Female	40	29	31	35-44	Low activity	3
0030	Female	23	29	87	Under 25	Budget enthusiasts	5
0031	Male	60	30	4	55+	Low activity	3
0032	Female	21	30	73	Under 25	Budget enthusiasts	5
0033	Male	53	33	4	45-54	Low activity	3
0034	Male	18	33	92	Under 25	Budget enthusiasts	5
0035	Female	49	33	14	45-54	Low activity	3
0036	Female	21	33	81	Under 25	Budget enthusiasts	5
0037	Female	42	34	17	35-44	Low activity	3
0038	Female	30	34	73	25-34	Budget enthusiasts	5
0039	Female	36	37	26	35-44	Low activity	3

Salary means annual income in k\$. Spend means spending score from 1 to 100.

Customer Data

Rows 40-78 of 200

ID	Gender	Age	Salary	Spend	Age Group	Segment	Cluster
0040	Female	20	37	75	Under 25	Budget enthusiasts	5
0041	Female	65	38	35	55+	Low activity	3
0042	Male	24	38	92	Under 25	Budget enthusiasts	5
0043	Male	48	39	36	45-54	Low activity	3
0044	Female	31	39	61	25-34	Budget enthusiasts	5
0045	Female	49	39	28	45-54	Low activity	3
0046	Female	24	39	65	Under 25	Budget enthusiasts	5
0047	Female	50	40	55	45-54	Balanced shoppers	5
0048	Female	27	40	47	25-34	Balanced shoppers	5
0049	Female	29	40	42	25-34	Balanced shoppers	5
0050	Female	31	40	42	25-34	Balanced shoppers	5
0051	Female	49	42	52	45-54	Balanced shoppers	5
0052	Male	33	42	60	25-34	Active shoppers	5
0053	Female	31	43	54	25-34	Balanced shoppers	5
0054	Male	59	43	60	55+	Active shoppers	5
0055	Female	50	43	45	45-54	Balanced shoppers	5
0056	Male	47	43	41	45-54	Balanced shoppers	5
0057	Female	51	44	50	45-54	Balanced shoppers	5
0058	Male	69	44	46	55+	Balanced shoppers	5
0059	Female	27	46	51	25-34	Balanced shoppers	5
0060	Male	53	46	46	45-54	Balanced shoppers	5
0061	Male	70	46	56	55+	Balanced shoppers	5
0062	Male	19	46	55	Under 25	Balanced shoppers	5
0063	Female	67	47	52	55+	Balanced shoppers	5
0064	Female	54	47	59	45-54	Balanced shoppers	5
0065	Male	63	48	51	55+	Balanced shoppers	5
0066	Male	18	48	59	Under 25	Balanced shoppers	5
0067	Female	43	48	50	35-44	Balanced shoppers	5
0068	Female	68	48	48	55+	Balanced shoppers	5
0069	Male	19	48	59	Under 25	Balanced shoppers	5
0070	Female	32	48	47	25-34	Balanced shoppers	5
0071	Male	70	49	55	55+	Balanced shoppers	5
0072	Female	47	49	42	45-54	Balanced shoppers	5
0073	Female	60	50	49	55+	Balanced shoppers	5
0074	Female	60	50	56	55+	Balanced shoppers	5
0075	Male	59	54	47	55+	Balanced shoppers	5
0076	Male	26	54	54	25-34	Balanced shoppers	5
0077	Female	45	54	53	45-54	Balanced shoppers	5
0078	Male	40	54	48	35-44	Balanced shoppers	5

Salary means annual income in k\$. Spend means spending score from 1 to 100.

Customer Data

Rows 79-117 of 200

ID	Gender	Age	Salary	Spend	Age Group	Segment	Cluster
0079	Female	23	54	52	Under 25	Balanced shoppers	5
0080	Female	49	54	42	45-54	Balanced shoppers	5
0081	Male	57	54	51	55+	Balanced shoppers	5
0082	Male	38	54	55	35-44	Balanced shoppers	5
0083	Male	67	54	41	55+	Balanced shoppers	5
0084	Female	46	54	44	45-54	Balanced shoppers	5
0085	Female	21	54	57	Under 25	Balanced shoppers	5
0086	Male	48	54	46	45-54	Balanced shoppers	5
0087	Female	55	57	58	55+	Balanced shoppers	5
0088	Female	22	57	55	Under 25	Balanced shoppers	5
0089	Female	34	58	60	25-34	Active shoppers	5
0090	Female	50	58	46	45-54	Balanced shoppers	5
0091	Female	68	59	55	55+	Balanced shoppers	5
0092	Male	18	59	41	Under 25	Balanced shoppers	5
0093	Male	48	60	49	45-54	Balanced shoppers	5
0094	Female	40	60	40	35-44	Balanced shoppers	5
0095	Female	32	60	42	25-34	Balanced shoppers	5
0096	Male	24	60	52	Under 25	Balanced shoppers	5
0097	Female	47	60	47	45-54	Balanced shoppers	5
0098	Female	27	60	50	25-34	Balanced shoppers	5
0099	Male	48	61	42	45-54	Balanced shoppers	5
0100	Male	20	61	49	Under 25	Balanced shoppers	5
0101	Female	23	62	41	Under 25	Balanced shoppers	5
0102	Female	49	62	48	45-54	Balanced shoppers	5
0103	Male	67	62	59	55+	Balanced shoppers	5
0104	Male	26	62	55	25-34	Balanced shoppers	5
0105	Male	49	62	56	45-54	Balanced shoppers	5
0106	Female	21	62	42	Under 25	Balanced shoppers	5
0107	Female	66	63	50	55+	Balanced shoppers	5
0108	Male	54	63	46	45-54	Balanced shoppers	5
0109	Male	68	63	43	55+	Balanced shoppers	5
0110	Male	66	63	48	55+	Balanced shoppers	5
0111	Male	65	63	52	55+	Balanced shoppers	5
0112	Female	19	63	54	Under 25	Balanced shoppers	5
0113	Female	38	64	42	35-44	Balanced shoppers	5
0114	Male	19	64	46	Under 25	Balanced shoppers	5
0115	Female	18	65	48	Under 25	Balanced shoppers	5
0116	Female	19	65	50	Under 25	Balanced shoppers	5
0117	Female	63	65	43	55+	Balanced shoppers	5

Salary means annual income in k\$. Spend means spending score from 1 to 100.

Customer Data

Rows 118-156 of 200

ID	Gender	Age	Salary	Spend	Age Group	Segment	Cluster
0118	Female	49	65	59	45-54	Balanced shoppers	5
0119	Female	51	67	43	45-54	Balanced shoppers	5
0120	Female	50	67	57	45-54	Balanced shoppers	5
0121	Male	27	67	56	25-34	Balanced shoppers	5
0122	Female	38	67	40	35-44	Balanced shoppers	5
0123	Female	40	69	58	35-44	Balanced shoppers	5
0124	Male	39	69	91	35-44	Active shoppers	1
0125	Female	23	70	29	Under 25	High income, low spend	2
0126	Female	31	70	77	25-34	Premium targets	1
0127	Male	43	71	35	35-44	High income, low spend	2
0128	Male	40	71	95	35-44	Premium targets	1
0129	Male	59	71	11	55+	High income, low spend	2
0130	Male	38	71	75	35-44	Premium targets	1
0131	Male	47	71	9	45-54	High income, low spend	2
0132	Male	39	71	75	35-44	Premium targets	1
0133	Female	25	72	34	25-34	High income, low spend	2
0134	Female	31	72	71	25-34	Premium targets	1
0135	Male	20	73	5	Under 25	High income, low spend	2
0136	Female	29	73	88	25-34	Premium targets	1
0137	Female	44	73	7	35-44	High income, low spend	2
0138	Male	32	73	73	25-34	Premium targets	1
0139	Male	19	74	10	Under 25	High income, low spend	2
0140	Female	35	74	72	35-44	Premium targets	1
0141	Female	57	75	5	55+	High income, low spend	2
0142	Male	32	75	93	25-34	Premium targets	1
0143	Female	28	76	40	25-34	Balanced shoppers	2
0144	Female	32	76	87	25-34	Premium targets	1
0145	Male	25	77	12	25-34	High income, low spend	2
0146	Male	28	77	97	25-34	Premium targets	1
0147	Male	48	77	36	45-54	High income, low spend	2
0148	Female	32	77	74	25-34	Premium targets	1
0149	Female	34	78	22	25-34	High income, low spend	2
0150	Male	34	78	90	25-34	Premium targets	1
0151	Male	43	78	17	35-44	High income, low spend	2
0152	Male	39	78	88	35-44	Premium targets	1
0153	Female	44	78	20	35-44	High income, low spend	2
0154	Female	38	78	76	35-44	Premium targets	1
0155	Female	47	78	16	45-54	High income, low spend	2
0156	Female	27	78	89	25-34	Premium targets	1

Salary means annual income in k\$. Spend means spending score from 1 to 100.

Customer Data

Rows 157-195 of 200

ID	Gender	Age	Salary	Spend	Age Group	Segment	Cluster
0157	Male	37	78	1	35-44	High income, low spend	2
0158	Female	30	78	78	25-34	Premium targets	1
0159	Male	34	78	1	25-34	High income, low spend	2
0160	Female	30	78	73	25-34	Premium targets	1
0161	Female	56	79	35	55+	High income, low spend	2
0162	Female	29	79	83	25-34	Premium targets	1
0163	Male	19	81	5	Under 25	High income, low spend	2
0164	Female	31	81	93	25-34	Premium targets	1
0165	Male	50	85	26	45-54	High income, low spend	2
0166	Female	36	85	75	35-44	Premium targets	1
0167	Male	42	86	20	35-44	High income, low spend	2
0168	Female	33	86	95	25-34	Premium targets	1
0169	Female	36	87	27	35-44	High income, low spend	2
0170	Male	32	87	63	25-34	Premium targets	1
0171	Male	40	87	13	35-44	High income, low spend	2
0172	Male	28	87	75	25-34	Premium targets	1
0173	Male	36	87	10	35-44	High income, low spend	2
0174	Male	36	87	92	35-44	Premium targets	1
0175	Female	52	88	13	45-54	High income, low spend	2
0176	Female	30	88	86	25-34	Premium targets	1
0177	Male	58	88	15	55+	High income, low spend	2
0178	Male	27	88	69	25-34	Premium targets	1
0179	Male	59	93	14	55+	High income, low spend	2
0180	Male	35	93	90	35-44	Premium targets	1
0181	Female	37	97	32	35-44	High income, low spend	2
0182	Female	32	97	86	25-34	Premium targets	1
0183	Male	46	98	15	45-54	High income, low spend	2
0184	Female	29	98	88	25-34	Premium targets	1
0185	Female	41	99	39	35-44	High income, low spend	4
0186	Male	30	99	97	25-34	Premium targets	1
0187	Female	54	101	24	45-54	High income, low spend	2
0188	Male	28	101	68	25-34	Premium targets	1
0189	Female	41	103	17	35-44	High income, low spend	2
0190	Female	36	103	85	35-44	Premium targets	1
0191	Female	34	103	23	25-34	High income, low spend	2
0192	Female	32	103	69	25-34	Premium targets	1
0193	Male	33	113	8	25-34	High income, low spend	2
0194	Female	38	113	91	35-44	Premium targets	1
0195	Female	47	120	16	45-54	High income, low spend	4

Salary means annual income in k\$. Spend means spending score from 1 to 100.

Customer Data

Rows 196-200 of 200

ID	Gender	Age	Salary	Spend	Age Group	Segment	Cluster
0196	Female	35	120	79	35-44	Premium targets	4
0197	Female	45	126	28	45-54	High income, low spend	4
0198	Male	32	126	74	25-34	Premium targets	4
0199	Male	32	137	18	25-34	High income, low spend	4
0200	Male	30	137	83	25-34	Premium targets	4